

Project 2 – Carolina Cougars

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1 Introduction

The National Basketball Association (NBA) is preparing its 2025–2026 regular-season schedule. Constructing such a schedule requires assigning every game to a specific date, location, and home–away pairing while satisfying a wide range of league rules and operational constraints. To support this process, a preliminary draft schedule has been provided. Although not optimized, it includes all essential structural information—game dates, participating teams, arenas, and start times—allowing us to analyze feasibility and reconstruct the schedule using mathematical modeling.

The objective of this project is to extract structural patterns from the draft schedule and formulate Integer Programming (IP) models that reproduce its feasible structure, incorporate additional operational considerations, and evaluate improved scheduling alternatives.

2 Background

2.1 NBA Scheduling as a Structured Optimization Problem

NBA scheduling is a classic large-scale combinatorial optimization problem. Each team must play 82 regular-season games, divided according to league-mandated matchup frequencies: four games against each division opponent, three–four games against remaining conference opponents, and two games against every team in the opposite conference. These fixed matchup requirements form strict combinatorial constraints where each assignment influences multiple others.

2.2 Operational Constraints

Beyond matchup frequencies, real NBA scheduling must account for balanced distribution of home and away games, limits on back-to-backs, avoiding long road trips or long homestands, arena availability constraints, and travel fairness and rest considerations. These factors increase the complexity of constructing a valid schedule.

2.3 Time-Zone and Travel Considerations

Teams routinely travel across multiple time zones, and empirical research shows that repeated or large time-zone transitions can negatively affect player performance and recovery. For the 2025–2026 season, an additional complication arises from the U.S. daylight-saving transition on November 2, 2025: all teams except Phoenix shift forward by one hour; after this date, time-zone differences involving Phoenix increase by one hour. This affects the cumulative time-zone changes between consecutive games and must be incorporated when evaluating scheduling constraints related to travel fatigue.

2.4 Integer Programming in Sports Scheduling

Sports scheduling is NP-hard, making IP a natural framework for modeling the NBA’s structured rules. Binary variables can encode matchup–date assignments, and linear constraints capture opponent frequencies, home/away patterns, and sequencing rules. Professional leagues often use hybrid methods combining IP, constraint programming, and metaheuristics. This project focuses on IP formulations to analyze and extend the draft schedule.

3 Part 1 – Data Analysis

3.1 Tasks

Tasks: a) For each team i , list all dates when i played at home. b) For each team j , compute how many times i played against j at home. c) For each team j , compute how many times i played against j away. d) For each team i , list all dates when i played away.

Notes: Include tables summarizing home/away dates and pairwise match counts. Mention how data were extracted or computed (e.g., using Python or pandas).

3.2 Objective of the Task

The goal of Problem 1 is to extract key structural features from the preliminary NBA 2025–2026 schedule. These extracted properties define the fixed components of the schedule and will later serve as non-negotiable constraints in the optimization model developed in Problem 2. In particular, for each team in the league, we determine the dates on which the team plays at home or away and the frequencies of matchups against every opponent.

3.3 Data Source

The analysis is based on the file `games.csv`, which contains the full preliminary schedule. Each row represents one NBA match and includes the following relevant fields:

- **Date** – The scheduled date of the match.
- **Home** – The team hosting the game.
- **Away** – The team playing on the road.

This dataset serves as the sole source of information for extracting structural schedule patterns.

3.4 Data Preparation

Before computing summary statistics, the dataset underwent minimal preprocessing to ensure consistency:

- **Column name standardization:** to avoid incompatibility or ambiguity, column names were checked and normalized. For example, if a version of the dataset used “Visitor” instead of “Away”, it was renamed to the required format.
- **Date format conversion:** the Date field was converted into a standardized datetime format using a parsing step. This enabled proper chronological sorting and eliminated formatting inconsistencies (e.g., converting variations such as “Oct 23, 2025” to a uniform date structure).

No missing data or duplicated match records were detected.

3.5 Methodology

After the dataset was cleaned and prepared, the next step was to extract structured scheduling information for each team. The method followed a systematic process. First, the set of all teams was identified by combining unique entries from both the Home and Away columns. This ensured that every team appearing in the schedule would be included in the analysis.

For each team, two subsets of matches were created. One subset contained all games where the team played at home, and the other contained games where the team played away. These subsets were then sorted chronologically using the parsed date column in order to preserve the original timing of matches throughout the season.

Once the ordering was established, frequency counts were calculated to determine how many times each team played against every opponent in both home and away settings. This allowed the schedule to be summarized not only in terms of game timing but also opponent distribution patterns.

Finally, the extracted results were stored in a structured format and exported as CSV and JSON files. These outputs will serve as direct parameter inputs for the optimization model constructed in the next part of the project.

3.6 Results

The analysis produced a structured summary of the preliminary schedule for all teams. For each team, four types of information were extracted: the dates of home games, the dates of away games, the number of times each opponent was hosted, and the number of times each opponent was played on the road. The results cover every team in the dataset and are organized consistently across all outputs.

The extracted data confirm that each team has a predefined pattern of home and away appearances and fixed matchup frequencies. These structural elements serve as immutable constraints in the optimization model of Problem 2.

To facilitate further processing, all results were exported into five files:

Output File	Description
q1_home_dates.csv	Home game dates for each team
q1_away_dates.csv	Away game dates for each team
q1_home_counts.csv	Home matchup frequencies against each opponent
q1_away_counts.csv	Away matchup frequencies against each opponent
q1_summary.json	Complete structured dataset for modeling input

Table 1: Problem 1 output files.

A representative example is shown below for the Atlanta Hawks. The team plays ten home games and six away games in the preliminary schedule. Each home game is against a unique opponent, while away games are distributed across six different opponents. The extracted schedule fragment is:

Representative Example: Atlanta Hawks

This example illustrates how the data capture both the temporal structure and the opponent distribution pattern for each team. Due to the volume of output, only a representative example is shown in this section, while the full results for all teams have been placed in the Appendix.

Table 2: Atlanta Hawks — Home Games

Game	Date	Opponent	Count
1	Mon, Nov 03, 2025	Boston Celtics	1
2	Fri, Nov 07, 2025	Chicago Bulls	1
3	Sat, Nov 15, 2025	Dallas Mavericks	1
4	Mon, Nov 17, 2025	Golden State Warriors	1
5	Wed, Nov 19, 2025	Los Angeles Lakers	1
6	Sun, Nov 23, 2025	Miami Heat	1
7	Thu, Nov 27, 2025	Milwaukee Bucks	1
8	Fri, Nov 28, 2025	New York Knicks	1
9	Sat, Nov 29, 2025	Phoenix Suns	1
10	Thu, Dec 25, 2025	Toronto Raptors	1

Table 3: Atlanta Hawks — Away Games

Game	Date	Opponent	Count
1	Sat, Nov 01, 2025	Brooklyn Nets	1
2	Wed, Nov 05, 2025	Chicago Bulls	1
3	Tue, Nov 11, 2025	Cleveland Cavaliers	1
4	Thu, Nov 13, 2025	Denver Nuggets	1
5	Fri, Nov 21, 2025	Houston Rockets	1
6	Mon, Dec 01, 2025	Philadelphia 76ers	1

From the full analysis of the schedule, we also confirm that no pair of teams played each other more than once, neither at home nor away. This follows from the output of the code, which shows exactly one encounter for every team pair whenever a matchup occurs.

Table 4: Summary of home and away dates for each team

Team	# Count Home Games	# Count Away Games
Atlanta Hawks	10	6
Boston Celtics	6	10
Brooklyn Nets	7	9
Chicago Bulls	9	7
Cleveland Cavaliers	10	6
Dallas Mavericks	9	7
Denver Nuggets	8	8
Golden State Warriors	7	9
Houston Rockets	9	7
Los Angeles Lakers	7	9
Miami Heat	8	8
Milwaukee Bucks	9	7
New York Knicks	6	10
Philadelphia 76ers	7	9
Phoenix Suns	8	8
Toronto Raptors	8	9

4 Part 2 – Feasibility Integer Program

4.1 Symbols Dictionary

Symbol	Description
T	Set of teams
D	Set of dates
H_i	Set of home dates for team i
A_i	Set of away dates for team i
x_{ijd}	1 if team i hosts team j on date d , 0 otherwise

Table 5: Sets, dates, and decision variable for Part 2.

4.2 Objective of the Task

The purpose of Problem 2 is to construct an Integer Programming (IP) formulation whose feasible region represents all possible schedules that preserve the structural properties extracted in Problem 1. The model does not aim to optimize or modify the schedule but rather ensures that any feasible solution strictly respects the original home/away structure and matchup frequencies recorded from the preliminary dataset.

This step establishes the mathematical foundation needed for later iterations in Problem 3 and later improvements, where feasibility will be tested under additional logistical constraints and optimization objectives.

4.3 Data Mapping and Model Parameters

To ensure consistency between the mathematical formulation and the preliminary schedule dataset, the raw game records were programmatically transformed into the structured parameters required by the integer programming model. The extraction process was automated to avoid manual transcription errors and guarantee that the model reflects the true schedule structure presented in the input file.

First, all unique teams and match dates appearing in the dataset were identified and stored as the sets T and D , respectively. The dataset was then analyzed to determine, for each team i , the exact dates on which the team appeared as the home team and as the away team. These results formed the sets H_i (home dates of team i) and A_i (away dates of team i), directly corresponding to the home and away patterns computed in Problem 1.

Next, the frequency of matchups between every pair of distinct teams was computed. For each ordered pair (i, j) , the number of times that team i hosted team j was recorded as parameter H_{ij} , and the number of times team i played away at team j was stored as A_{ij} . These counts ensure that the integer programming model reproduces the exact opponent frequencies contained in the original schedule.

The output of this mapping process is a complete and consistent set of model parameters $\{T, D, H_i, A_i, H_{ij}, A_{ij}\}$, which serve as the foundation for the constraint system established in the subsequent section. The automated extraction guarantees full alignment between the model input and the real schedule information, thereby ensuring that any feasible solution to the model corresponds to a structurally valid version of the preliminary schedule.

A brief tabular summary of the matchup-frequency parameters is given in Table 6.

Symbol	Description
H_{ij}	Number of times team i hosts team j
A_{ij}	Number of times team i plays away at team j

Table 6: Matchup frequency parameters.

4.4 Decision Variable Construction

Following the generation of model parameters, the next step involves defining the decision variables that represent the potential match scheduling assignments within the integer programming formulation. The decision variable is modeled as a binary indicator x_{ijd} , defined as:

$$x_{ijd} = \begin{cases} 1 & \text{if team } i \text{ plays at home vs. team } j \text{ on day } d, \\ 0 & \text{otherwise.} \end{cases}$$

To ensure that the model structure reflects realistic scheduling logic and avoids infeasible or meaningless combinations, variables are created only when all of the following conditions hold:

- Team i is scheduled as a home team on date d ;
- Team j is scheduled as an away team on the same date;
- The pair (i, j) appears at least once in the original schedule dataset.

This filtering strategy significantly reduces the dimensionality of the model by excluding variable combinations that cannot occur under the original constraints of the schedule. As a result, the model remains computationally efficient while preserving the full structural fidelity of the preliminary calendar.

The resulting decision variable set guarantees that every binary choice encoded in the optimization model corresponds to a feasible match configuration in practice. This ensures alignment between data and modeling logic, and it provides the foundation for enforcing the structural feasibility constraints introduced in the next section.

4.5 Mathematical Model

The mathematical formulation developed in this step represents all feasible schedules that comply with the structural properties extracted from the preliminary dataset. The model operates exclusively on the decision variable defined in Section 4.4 and uses the parameter sets derived in Section 4.3. Since the objective at this stage is feasibility verification rather than optimization, no objective function is included, and the model is solved as a pure feasibility integer program. The constraints (e) Home dates fixed to H_i . (f) Away dates fixed to A_i . (g) Number of home matches vs. each team j . (h) Number of away matches vs. each team j . Plus additional logical constraints (no overlapping games, etc.), are written in mathematical form as follow:

(e) Home dates fixed to H_i :

$$\sum_{j \in T \setminus \{i\}} x_{ijd} = 1 \quad \forall i \in T, \forall d \in H_i.$$

(f) Away dates fixed to A_i :

$$\sum_{j \in T \setminus \{i\}} x_{jid} = 1 \quad \forall i \in T, \forall d \in A_i.$$

(g) **Number of home matches vs. each team j :**

$$\sum_{d \in D} x_{ijd} = H_{ij} \quad \forall i, j \in T, i \neq j.$$

(h) **Number of away matches vs. each team j :**

$$\sum_{d \in D} x_{jid} = A_{ij} \quad \forall i, j \in T, i \neq j.$$

Additional logical constraint: no overlapping games for a team on the same date:

$$\sum_{j \in T \setminus \{i\}} (x_{ijd} + x_{jid}) \leq 1 \quad \forall i \in T, \forall d \in D.$$

Binary domain:

$$x_{ijd} \in \{0, 1\} \quad \forall i, j \in T, \forall d \in D.$$

4.6 Implementation and Feasibility Check

To verify whether a feasible schedule exists under the constraints defined in the mathematical model, the formulation was implemented and executed using the Gurobi optimization solver in Python. The parameter sets derived in Section 4.3 and the decision variables defined in Section 4.4 were passed into the solver, where the full constraint system from Section 4.5 was applied.

Since this stage focuses exclusively on feasibility rather than optimization, the model was solved without an objective function, and the solver was instructed to search for any solution satisfying all imposed constraints. If such a solution exists, it confirms that the extracted structure from the original schedule is logically consistent and can be reconstructed within the defined scheduling framework. Conversely, an infeasibility result would indicate structural inconsistency within the preliminary schedule or the imposed constraint set.

The implementation also includes an internal mechanism for validating variable indexing, ensuring that only logically possible scheduling combinations are created and evaluated. Once the model was executed, the solver status was used to determine whether the schedule is feasible. A feasible solution, if identified, will be used as the baseline for further analysis in subsequent steps of the project.

4.7 Results

After implementing the formulation in Gurobi and executing the feasibility check, the solver successfully identified a feasible solution that satisfies all imposed constraints. The model was reduced during presolve, indicating that redundant or logically implied constraints and variables were removed, which improved computational efficiency. The final reduced model consisted of 308 constraints and 442 binary variables.

The solver determined feasibility without requiring branching, as indicated by the fact that zero branch-and-bound nodes were explored. A valid solution was found during the heuristic phase, and the solver confirmed optimality with a zero objective value, which is expected since the model was formulated as a feasibility problem without an optimization objective.

The solver output concluded with: *“Optimal solution found ... Schedule feasible.”*

This result confirms that the structural properties extracted from the preliminary dataset are mathematically consistent and allow for at least one complete schedule that satisfies all home/away patterns and matchup frequency requirements. Therefore, the model can serve as a valid foundation for subsequent steps in the project, such as incorporating additional logistical constraints or pursuing optimization objectives.

5 Part 3 – Additional Time-Zone Constraint

Time-zone information for all NBA arenas was obtained from an online world clock resource.¹ A key correction concerns the U.S. Daylight Saving Time change on November 2, 2025: all arenas shifted their UTC offset by one hour except Phoenix, which does not observe DST. This causes Phoenix’s time-zone difference relative to all other teams to increase by one hour after the first match. The adjustment was incorporated directly into the dataset and implemented in the Python code used for the model.

5.1 Objective

The purpose of this problem is to extend the feasible scheduling model from Part 2 by introducing a travel-fatigue rule based on NBA arena time zones. The NBA spans multiple time zones (Eastern, Central, Mountain, Pacific), and long repeated jumps across zones can cause unrealistic travel patterns for teams.

We impose the following league-style constraint:

No team is allowed to play three consecutive games for which the sum of the time-zone changes between consecutive games is 4 hours or more.

Formally, for three consecutive games g_1, g_2, g_3 played by the same team:

$$|TZ(g_2) - TZ(g_1)| + |TZ(g_3) - TZ(g_2)| < 4,$$

where $TZ(g)$ is the UTC offset (in integer hours) of the arena where game g is played.

This ensures that sequences containing, for example, any 2-hour jump followed by a 3-hour jump are prohibited. This problem checks whether a schedule that is structurally identical to the original one can also satisfy this fatigue constraint.

5.2 Sets, Indices, and Parameters

Table 7: Sets, indices, and parameters for the time-zone constraint model.

Symbol	Description
$G = \{0, 1, \dots, G - 1\}$	Set of all games; elements indexed by g
T	Set of teams; elements indexed by t
$D = \{0, 1, \dots, D - 1\}$	Chronological date slots; elements indexed by d
$\text{Home}(g)$	Home team of game g (takes values in T)
$\text{Away}(g)$	Away team of game g (takes values in T)
$TZ(g)$	UTC offset of the arena where game g is played (used to set z_{td} when team t plays)
M	Big-M constant used to deactivate triple-day constraints when a team does not play on all three dates ($M = 20$)

5.3 Decision Variables

We introduce the following decision variables, all defined consistently with the game–day assignment framework used in Part 3.

¹Time and Date AS, “World Clock,” available at <https://www.timeanddate.com/worldclock/>, accessed on December 1, 2025.

- **Primary assignment variable**

$$x_{gd} = \begin{cases} 1 & \text{if game } g \text{ is scheduled on date } d, \\ 0 & \text{otherwise.} \end{cases}$$

This binary variable assigns each game to exactly one date and guarantees that no game is played more than once.

- **Team-play indicator**

$$y_{td} = \begin{cases} 1 & \text{if team } t \text{ plays on date } d, \\ 0 & \text{otherwise.} \end{cases}$$

This variable is implied by the game assignment but is introduced explicitly to simplify constraints involving team-specific patterns, such as consecutive-day time-zone transitions.

- **Team time-zone variable**

$$z_{td} \in \mathbb{R},$$

representing the effective time-zone offset of team t on date d . When a team plays, this value matches the UTC offset of the arena where the game takes place. When the team does not play, the variable is unconstrained (but rendered irrelevant through big- M logic).

- **Time-zone difference variables** For each team t and each triple of consecutive dates $(d, d+1, d+2)$, we introduce the absolute time-zone jumps:

$$a_{td}^{(1)} = |z_{t,d+1} - z_{t,d}|, \quad a_{td}^{(2)} = |z_{t,d+2} - z_{t,d+1}|.$$

These variables represent directly the magnitudes of consecutive time-zone transitions.

5.4 Model Formulation

Before introducing the time-zone constraint, we recall that all feasibility requirements from Part 1 and Part 2 remain in force. In particular, the schedule must satisfy the fixed home/away dates, matchup frequencies, and non-overlapping conditions established in the feasibility model.

In addition to these structural constraints, we now introduce an additional constraint designed to limit excessive travel fatigue due to large time-zone transitions. Using the absolute jump variables defined above, we impose the following condition: for each team t and each triple of consecutive dates $(d, d+1, d+2)$, the total time-zone change across the two transitions must remain strictly below four hours:

$$a_{td}^{(1)} + a_{td}^{(2)} < 4.$$

This ensures that no team experiences an excessive cumulative time-zone shift over any three-game sequence.

As in Part 2, we do not specify an objective function; the purpose of this formulation is solely to determine whether a feasible schedule exists that satisfies all structural and time-zone constraints simultaneously.

5.5 Extracting the Feasible Schedule

After optimization, we reconstruct the schedule by:

- identifying all (g, d) pairs where the game–date assignment variable equals 1;
- converting date indices back to calendar dates;

- creating a final dataframe with: game index, date, visitor, home, arena, UTC offset.

The final schedule is exported as `final_schedule.csv`. This schedule satisfies:

- all structural constraints from Part 2;
- all time-zone constraints from Part 3.

5.6 Results

Using Integer Programming and solving the model with Gurobi, we obtained a feasible schedule that satisfies both the structural constraints inherited from Part 2 and the additional time-zone fatigue rule introduced in Part 3. The optimization is particularly sensitive to the early-November U.S. transition from Daylight Saving Time to Standard Time, so the model computes the effective UTC offset of every game based on the date to which it is assigned, rather than the original draft date. This ensures realistic handling of the DST switch.

The effect is especially noticeable for Phoenix, whose home arena in Arizona does not observe Daylight Saving Time and remains at UTC−7 all year. As a result, when the DST change occurs on November 2, all other U.S. arenas shift by one hour while Phoenix does not. The model correctly accounts for this: Phoenix’s relative offset changes after its first game not because Phoenix moved, but because the rest of the league shifted around it.

After solving, we validated every team’s sequence of consecutive games using these date-adjusted offsets. In all cases, the sum of the consecutive time-zone jumps remained strictly below 4, confirming that the Gurobi solution fully satisfies the fatigue constraint and handles DST—including the Arizona exception—correctly.

5.7 Results

After incorporating the time-zone fatigue rule into the scheduling model, we attempted to solve the extended integer program under the original requirement

$$|TZ_2 - TZ_1| + |TZ_3 - TZ_2| < 4.$$

Gurobi returned INFEASIBLE, indicating that the structural constraints from Part 2—namely the fixed home/away dates and the predetermined calendar of games—cannot be satisfied simultaneously with such a strict fatigue bound.

A detailed examination of the ordered date structure (`DateSlot`) and arena time-zone offsets (`ArenaUTCOffset`) revealed that several teams are *structurally forced*, due to fixed home/away patterns, into unavoidable long-distance three-game sequences. These sequences include cross-country road trips and, notably, stretches spanning the early-November U.S. daylight-saving transition, during which most arenas shift from UTC−4 to UTC−5, while Phoenix remains at UTC−7 throughout. Because these transitions occur on fixed calendar dates, no opponent reassignment can eliminate cumulative jumps of magnitude ≥ 4 hours, rendering the model mathematically infeasible under the original threshold.

To identify the tightest feasible bound, we progressively relaxed the time-zone limit and resolved the model. The solver remained INFEASIBLE for all integer thresholds below 6, but returned OPTIMAL once the constraint was relaxed to

$$|TZ_2 - TZ_1| + |TZ_3 - TZ_2| < 6.$$

This establishes 6 hours as the *minimal feasible fatigue threshold* consistent with the league’s fixed calendar and home/away structure. In other words, a complete schedule can be produced only when the cumulative three-game time-zone change is allowed to reach, but not attain, 6 hours.

The resulting optimal schedule satisfying the 6-hour fatigue rule has been reconstructed and included with our submission as **Schedule Part 3-4.csv**. This schedule is fully consistent with all structural NBA-style requirements and imposes the maximally strict feasible fatigue constraint given the part 2 and 3 restrictions.

6 Part 4 – Scheduling Improvement

6.1 Process

6.2 Additional Mathematical Considerations for the Distance-Based Model

The distance-minimization model extends the feasibility structure introduced in Part 2. All structural constraints from the original formulation are preserved: every team must play on the same dates, with the same home/away pattern, and no team may play more than one game per day. The main modification concerns opponent assignment: unlike Part 2, the visiting team on each date is no longer fixed. The model is instead allowed to reconfigure the home–visitor pairings while strictly maintaining the total number of home and road games for every team. All remaining feasibility constraints from earlier sections remain unchanged.

Geographical coordinates (latitude and longitude) for each team’s home arena were obtained from a publicly available geolocation resource,² and are used to compute great-circle distances between cities via the haversine formula. These distances define the cost associated with any potential road game in the optimized schedule.

Mathematically, the model introduces a set of variables and parameters specifically designed to evaluate travel distance and to measure fairness across teams. The relevant symbols used in this section are summarized in Table 8.

Table 8: Parameters and decision variables for the distance-minimization model.

Symbol	Description
$y_{ijd} \in \{0, 1\}$	Binary variable equal to 1 if team i plays at home against team j on date d .
Dist_t	Total travel distance of team t in the optimized schedule (sum of distances when t is the away team).
dist_{ij}	Great-circle distance between the home cities of teams i and j , computed from their GPS coordinates.
μ	Mean travel distance across all teams in the optimized schedule: $\mu = \frac{1}{ T } \sum_{t \in T} \text{Dist}_t$.
$\text{dev}_t^+, \text{dev}_t^- \geq 0$	Positive and negative deviations such that $\text{Dist}_t - \mu = \text{dev}_t^+ - \text{dev}_t^-$, used to model absolute deviations.
A	Weighting parameter (set to $A = 5$) used to amplify the fairness component of the objective function.

The objective function combines two components: (i) the league-wide average distance, and (ii) an L1 fairness term that penalizes deviations from this average. Formally, the objective minimized is

$$\min(\mu + A \sum_{t \in T} |\text{Dist}_t - \mu|),$$

²GPS-Coordinates.net, “Find GPS Coordinates,” available at <https://www.gps-coordinates.net/>, accessed on December 1, 2025.

where $A = 50$ is a scaling parameter introduced to strengthen the fairness component. This choice is deliberate: the L1 deviation term measures how unequal the travel burden is across teams, and increasing its weight reduces the presence of outliers—teams that would otherwise enjoy disproportionately low travel or bear excessively high travel obligations.

This approach prioritizes a more equitable travel distribution over purely minimizing total distance, aligning the scheduling design with a fairness-oriented perspective.

6.3 Results

To evaluate potential improvements to the feasible schedule obtained in Question 2 and Question 3, we redesigned the game assignment order while preserving all original structural constraints.

This ensures that all feasibility conditions established in the previous sections remain valid, and only the **sequence of games** is reshuffled to reduce travel-related burden.

Sanity Check of Results

A structural validation was performed to verify that the optimized schedule maintains the home/away structure of the original schedule. The comparison table below confirms that for each team, both home and away counts remain identical to the baseline (all entries marked **TRUE**). This validates that the optimization is not altering the fundamental season design but only rearranging match order.

Table 9: Home/Visitor matchup consistency check before and after optimization

Team	Visitor New	Visitor Old	Home New	Home Old	Visitor Match?	Home Match?
Cleveland Cavaliers	6	6	10	10	TRUE	TRUE
Dallas Mavericks	7	7	9	9	TRUE	TRUE
Los Angeles Lakers	9	9	7	7	TRUE	TRUE
Houston Rockets	7	7	9	9	TRUE	TRUE
Golden State Warriors	9	9	7	7	TRUE	TRUE
Phoenix Suns	8	8	8	8	TRUE	TRUE
Atlanta Hawks	6	6	10	10	TRUE	TRUE
Denver Nuggets	8	8	8	8	TRUE	TRUE
Philadelphia 76ers	9	9	7	7	TRUE	TRUE
Toronto Raptors	8	8	8	8	TRUE	TRUE
Boston Celtics	10	10	6	6	TRUE	TRUE
Brooklyn Nets	9	9	7	7	TRUE	TRUE
Milwaukee Bucks	7	7	9	9	TRUE	TRUE
Miami Heat	8	8	8	8	TRUE	TRUE
New York Knicks	10	10	6	6	TRUE	TRUE
Chicago Bulls	7	7	9	9	TRUE	TRUE

6.4 Travel Distance Outcomes

Using geographic arena coordinates, we computed the total accumulated travel distance for each team before and after optimization. Table 10 summarizes the results.

The global improvement is measured using the mean travel distance and a fairness metric:

Original mean distance = 15417.94 km

Optimized mean distance = 11680.66 km

Reduction $\approx 24.2\%$

Mean absolute distance difference = 4602.87 km

Table 10: Travel distance comparison before and after optimization (updated results)

Team	Original (km)	Optimized (km)	Difference (km)
Atlanta Hawks	7113.70	8381.76	+1268.06
Boston Celtics	18964.26	10618.31	-8345.95
Brooklyn Nets	15840.41	11230.15	-4610.26
Chicago Bulls	9191.33	11357.52	+2166.19
Cleveland Cavaliers	12347.77	6480.54	-5867.23
Dallas Mavericks	12666.82	10496.86	-2169.96
Denver Nuggets	13229.78	11817.87	-1411.91
Golden State Warriors	26665.97	22135.73	-4530.24
Houston Rockets	11380.94	11619.56	+238.61
Los Angeles Lakers	25808.31	15639.60	-10168.71
Miami Heat	15117.61	11676.66	-3440.94
Milwaukee Bucks	8375.40	11627.37	+3251.97
New York Knicks	21323.36	11438.27	-9885.13
Philadelphia 76ers	17495.26	8055.63	-9439.63
Phoenix Suns	20380.72	14185.36	-6195.36
Toronto Raptors	10785.32	10129.48	-655.84

Original variance = 4819.09

New variance = 2131.99

Reduction $\approx 55.8\%$

These results confirm that the optimized schedule significantly reduces both total travel requirements and the inequality in travel distribution across teams. The new schedule not only lowers the average travel distance but also substantially reduces the disparity of travel demand between teams.

6.5 Interpretation and Discussion

The optimized schedule leads to improvements along two key dimensions:

- **Efficiency:** league-wide average team travel is significantly reduced, lowering fatigue and logistical cost.
- **Fairness:** distance variance drops by more than half, meaning no team is overburdened relative to others.

Teams originally facing extremely long travel schedules (e.g., Lakers, Knicks, Suns) benefited notably from the optimization, whereas a small number of teams experienced slight increases to balance overall fairness. The resulting travel distribution is therefore more equitable while remaining fully feasible.

It is important to note that our model intentionally prioritizes **fairness over pure efficiency**. Rather than minimizing total travel distance at all costs, the formulation is designed to avoid granting structural advantages to teams whose geographical position naturally reduces travel. By amplifying the fairness term in the objective function, the model promotes a more balanced distribution of travel load across the league, even if this comes at the expense of achieving the absolute minimum possible total distance.

6.6 Attachments

Link to the code: <https://github.com/dc3941-pixel/Project-2-Optimization> Reschedule with 6 hours difference as a threshold of Part 3 and Part 4: attached in the other submission link.